📎 SOFTWARE ENGINEERING – ASSESSMENT TOOL 1

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- **Problem No:** 36
- **Title:** Autonomous Electric Vehicle Charging Network Management System

🏭 INDUSTRY CONTEXT

- The transportation sector is rapidly adopting digital technologies to improve efficiency, safety, scalability, and decision making.
- Traditional approaches often rely on manual processes and reactive monitoring, resulting in delays, increased operational costs, resource wastage, and reduced service quality.
- There is a growing need for an intelligent, cloud-enabled, distributed platform capable of collecting real-time IoT and enterprise data, analyzing operational conditions, predicting risks, and supporting timely interventions.

🎯 PROBLEM STATEMENT

Design and develop a distributed Autonomous Electric Vehicle Charging Network Management System that continuously acquires real-time operational data from connected devices, enterprise systems, and users. The platform should detect abnormal events, generate predictive insights using AI/ML, notify stakeholders in real time, provide centralized dashboards, generate reports, and support secure, scalable, fault-tolerant operation across multiple locations.

📌 OBJECTIVES

- Collect real-time operational data.
- Monitor system performance continuously.
- Detect anomalies automatically.
- Predict failures or risks using analytics.
- Notify stakeholders immediately.
- Generate dashboards and reports.
- Improve operational efficiency.
- Ensure scalability, security, and high availability.

🌟 EXPECTED OUTCOMES

- Improved operational visibility.
- Reduced downtime and costs.
- Higher service quality and reliability.
- Faster decision making.
- Better resource utilization.
- Enhanced user/customer satisfaction.
- Support for large-scale deployment.

💡 PROPOSED SOLUTION

The proposed system is an intelligent, cloud-enabled distributed platform that gathers data from EV charging stations and enterprise systems. Using AI/ML algorithms, the system analyzes this data to predict risks and automatically detect abnormal events. It instantly notifies stakeholders and provides centralized dashboards to ensure seamless, fault-tolerant management across multiple large-scale locations.

⚙️ FUNCTIONAL REQUIREMENTS

- **Data Acquisition:** Continuous intake from IoT devices.
- **Anomaly Detection:** Automated flagging of abnormal events.
- **Predictive Analytics:** AI/ML-driven failure and risk prediction.
- **Alerting System:** Real-time stakeholder notifications.
- **Visualization:** Centralized dashboards and automated reporting.
- **Enterprise Integration:** Synchronization with existing user and enterprise systems.

🛡️ NON-FUNCTIONAL REQUIREMENTS

- **Performance:** Real-time operational data processing.
- **Availability:** Fault-tolerant and highly available architecture.
- **Security:** Encrypted data pipelines and secure stakeholder access.
- **Scalability:** Support for large-scale, multi-location deployment.
- **Maintainability:** Centralized updates via cloud infrastructure.

🔲 MODULES

1. **Real-Time Data Collection Engine**
2. **AI/ML Predictive Analytics Module**
3. **Anomaly Detection & Monitoring System**
4. **Stakeholder Notification & Alerting Hub**
5. **Dashboard & Report Generator**

👥 USERS

- **Network Administrator**
- **Maintenance Engineer**
- **EV Customer / User**
- **Operations Manager**

📥 INPUTS

IoT sensor data from charging stations (voltage, temperature, connection status), enterprise system data, user authentication details, and real-time operational metrics.

📤 OUTPUTS

Real-time automated alerts, centralized performance dashboards, predictive risk insights, operational efficiency reports, and anomaly logs.

💻 TECHNOLOGY STACK

- **Frontend:** HTML, CSS, JavaScript (React/Angular)
- **Backend:** Python/Node.js/Java
- **Database:** PostgreSQL/MongoDB
- **Cloud:** AWS/Azure for distributed deployment
- **AI/ML:** Machine Learning pipelines for predictive insights
- **IoT:** Connected charging device protocols (e.g., OCPP)

✅ ADVANTAGES

- Reduces operational costs and downtime.
- Eliminates resource wastage from manual monitoring.
- Optimizes resource utilization across locations.
- Drastically improves overall service quality and customer satisfaction.

🚀 FUTURE ENHANCEMENTS

- Integration with Vehicle-to-Grid (V2G) energy distribution.
- Dynamic AI-based pricing models.
- Advanced autonomous fleet charging support.
- Deep learning for advanced battery degradation forecasting.

🏁 CONCLUSION

The Autonomous Electric Vehicle Charging Network Management System addresses the critical need for digitalization in the transportation sector. By combining real-time IoT monitoring, AI/ML predictive analytics, and a cloud-enabled distributed architecture, the platform transitions the network from reactive monitoring to proactive management. This ultimately ensures high availability, reduces operational costs, and supports the scalable, large-scale deployment of electric vehicle infrastructure.